EMBEDDED SYSTEMS PROJECT

Snake Game on AVR Microcontroller

ATmega32 · 20×4 LCD · AVR-Embedded C

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System Architecture

5 PUSH BUTTONS
PA0–PA4 · input

RESET
state control · input

AVR ATmega32
8-bit MCU
Central control + game state

20×4 LCD
PORTD + PORTC

7-SEGMENT
PORTB · BCD score

BUZZER
PC3 · audio cue

Software Architecture (Layered Design)

APPLICATION LAYER

main.c · FUNCTION.c

HAL LAYER

LCD · Buzzer · Segment

MCAL LAYER

DIO

LIB LAYER

STD_TYPES · BIT_MATH · FUNCTION.h

BCD 7-Segment

0–99 · BCD OUTPUT

This part is responsible for displaying the game score on the 7-segment display. The score is represented using BCD (Binary-Coded Decimal), where each decimal digit is converted into its binary form before being sent to the display. The 7-segment is connected through PORTB

```c
void SevenSegment_Init(void) {

    MDIO_voidSetPortDirection(SEVEN_SEGMENT_PORT, 0xFF);
}

void SevenSegment_DisplayNumber(u8 NumberOfScore) {
    if (NumberOfScore <= 99) {
        u8 Units = NumberOfScore % 10;
        u8 Tens  = NumberOfScore / 10;

        u8 Score_Count = (Tens << 4) | Units;

        MDIO_voidSetPortValue(SEVEN_SEGMENT_PORT, Score_Count);
    }
}
```

BUZZER

This part controls the buzzer. The buzzer is connected to PC3, so the program can change the PC3 output to turn the buzzer on or off. It is mainly used to give the player an audio indication during the game

```
void HBuzzer_voidOn(void)
{
    MDIO_voidSetPinValue(
            BUZZER_PORT,
            BUZZER_PIN,
            DIO_HIGH);
}
```

```
void HBuzzer_voidOff(void)
{
    MDIO_voidSetPinValue(
            BUZZER_PORT,
            BUZZER_PIN,
            DIO_LOW);
}
```

```
void HBuzzer_voidBeep(void)
{
    HBuzzer_voidOn();

    _delay_ms(100);

    HBuzzer_voidOff();
}
```

```
void HBuzzer_voidInit(void)
{
    MDIO_voidSetPinDirection(
            BUZZER_PORT,
            BUZZER_PIN,
            DIO_OUTPUT);

    HBuzzer_voidOff();
}
```

LCD

This part controls the 20×4 LCD. The LCD is used to display the snake, food, score, and game messages. The program sends commands and data to the LCD to control what appears on the screen

```c
void HLCD_voidSendCommand(u8 A_u8Command);
void HLCD_voidSendData(u8 A_u8Command);
void HLCD_voidInit(void);
void HLCD_voidSendString(u8 *A_Pu8String);
void HLCD_voidClearDisplay(void);
void HLCD_voidDisplayNumberUnsigned (u32 A_u32Number);
void HLCD_voidDisplayNumberSigned (s32 A_s32Number);
void HLCD_voidGoToPos (LCD_ROWS A_LcdRowNo,LCD_COLS A_LcdColNo);
void HLCD_voidSendStringLine(u8 *A_Pu8String, LCD_ROWS row);

void HLCD_voidCreateCustomCharacters(void);
void HLCD_voidSendSpecialCharacter(u8 *A_pu8PatternArr, u8 A_u8PatternNumber);
```

```c
void HLCD_voidSendSpecialCharacter(u8 *A_pu8PatternArr, u8 A_u8PatternNumber)
{
    u8 local_u8CGRamAddress;

    // Calculate CGRAM Address = Pattern No. * 8
    local_u8CGRamAddress = A_u8PatternNumber * 8;

    // bit 6 must be high as it selects CGRAM to receive the next command
    SET_BIT(local_u8CGRamAddress, 6);

    // Send CGRAM Write Command
    HLCD_voidSendCommand(local_u8CGRamAddress);

    for(u8 i = 0; i < 7; i++)
    {
        HLCD_voidSendData(A_pu8PatternArr[i]);
    }

}
```

Custom LCD Characters

0 SOLID BLOCK

```
u8 solidBlock[7] = {
    0b01110,
    0b11111,
    0b11111,
    0b11111,
    0b11111,
    0b11111,
    0b01110
};
```

4 HEAD UP

```
u8 headUp[7] = {
    0b00000,
    0b00000,
    0b00000,
    0b00000,
    0b00100,
    0b01110,
    0b11111
};
```

3 HEAD LEFT

```
u8 headLeft[7] = {
    0b00001,
    0b00011,
    0b00111,
    0b01111,
    0b00111,
    0b00011,
    0b00001
};
```

2 HEAD RIGHT

```
u8 headRight[7] = {
    0b10000,
    0b11000,
    0b11100,
    0b11110,
    0b11100,
    0b11000,
    0b10000
};
```

1 DIAMOND FOOD

```
u8 diamond[7] = {
    0b00000,
    0b00100,
    0b01110,
    0b11111,
    0b01110,
    0b00100,
    0b00000

};
```

5 HEAD DOWN

```
u8 headDown[7] = {
    0b11111,
    0b01110,
    0b00100,
    0b00000,
    0b00000,
    0b00000,
    0b00000

};
```

This part defines custom characters for the game. Different character numbers are used for the snake body, food, and the four head directions. The program displays the required character depending on the snake's current state.

Game Loop Flowchart

MAIN

main() is the starting point of the program. It initializes the required hardware and then starts the game process. The main function connects the initialization stage with the main game loop

```
int main(void)
{
    gameInit();
    while(1) {

        startScreen();
        gameReset();
        u8 g_gameOver= gameLoop();

        if(!g_gameOver) //got out of gameloop without losing,then RESET button is hit
        {
            continue;
        }

        gameOverScreen();

        while(1)
        {
            if(READ_RESET() == DIO_LOW)
            {
            _delay_ms(200);
            break;
            }
        }

    }
    return 0;
}
```

MAIN DATA TYPES USED

This part defines the data types used by the game. These data types make it easier to store and handle information such as positions, directions, and game-related values.

```
static Snake g_snake; //struct holds all data related to snake
static Position g_food; //x-y coordinations of food

static u8 g_gameRunning; //flags
static u8 g_gameOver;

static u8 g_displayBuffer[4][20]; //all data to be sent to LCD

//initial value of button flags
static u8 g_prevUp = 1;
static u8 g_prevDown = 1;
static u8 g_prevLeft = 1;
static u8 g_prevRight = 1;
static u8 g_prevReset = 1;
```

```
typedef enum
{
    MOVE_UP=0,
    MOVE_DOWN,
    MOVE_RIGHT,
    MOVE_LEFT,
    RESET
} BUTTON_PINS;

typedef struct {
    u8 x;
    u8 y;
} Position;

typedef enum {
    DIR_UP = 0,
    DIR_DOWN,
    DIR_LEFT,
    DIR_RIGHT,
    DIR_NONE
} Direction;

//SORTED BY SIZE TO AVOID MEMORY LEAK
typedef struct {
    Position body[MAX_SNAKE_LENGTH];
    u16 score;
    Direction nextDir;
    Direction currentDir;
    u8 length;
    u8 isDead;
} Snake;
```

snakeInit

snakeInit() initializes the snake before gameplay starts. It sets the initial values needed for the snake, such as its starting position and initial state.

```
void snakeInit(void) //initializes data for each round
{
    //sets snake at the center of LCD ,directed to the right

    g_snake.length = 4;

    g_snake.currentDir = DIR_RIGHT;
    g_snake.nextDir = DIR_RIGHT;

    g_snake.isDead = 0;

    g_snake.score = 0;

    SevenSegment_DisplayNumber(0);

    g_snake.body[0].x = 8;
    g_snake.body[0].y = 1;

    g_snake.body[1].x = 7;
    g_snake.body[1].y = 1;

    g_snake.body[2].x = 6;
    g_snake.body[2].y = 1;

    g_snake.body[3].x = 5;
    g_snake.body[3].y = 1;


    HLCD_voidClearDisplay();

    generateFood();

    g_gameOver = 0;
    g_gameRunning = 1;

    g_prevUp = 1;
    g_prevDown = 1;
    g_prevLeft = 1;
    g_prevRight = 1;
    g_prevReset = 1;
}
```

Food functions

The food functions handle the food used in the game. They determine and manage the food position so that the snake can interact with it during gameplay

```
void generateFood(void) //generate random coordinates of food position
{
    do {
        g_food.x = rand() % 20;// random number range => 0:19
        g_food.y = rand() % 4;// random number range => 0:3
    } while(checkFoodPosition(g_food.x, g_food.y));
}

u8 checkFoodPosition(u8 x, u8 y)// checks if the random position is on the snake body
{
    for(u8 i = 0; i < g_snake.length; i++)
    {
        if(g_snake.body[i].x == x && g_snake.body[i].y == y)
        {
            return 1;
        }
    }

    return 0;
}
```

button input

This part reads the push-button inputs. The buttons are used by the player to control the snake's direction. The program checks which button is pressed and uses this input to control the game.

```c
void buttonInput(void) //handle input of buttons
{
    // checks falling edge

    u8 currentUp = READ_UP();
    u8 currentDown = READ_DOWN();
    u8 currentLeft = READ_LEFT();
    u8 currentRight = READ_RIGHT();
    u8 currentReset = READ_RESET();


    if(IS_PRESSED(currentReset, g_prevReset))
    {
        g_gameRunning=0;
        _delay_ms(200);
        g_prevReset = currentReset;

        return;
    }

    // prevent 180-direction change
    if(IS_PRESSED(currentUp, g_prevUp))
    {
        if(g_snake.currentDir != DIR_DOWN)
        {
            g_snake.nextDir = DIR_UP;
        }
    }
```

```c
#define READ_UP()     (MDIO_pinValueGetPinValue(BUTTON_PORT, MOVE_UP))
#define READ_DOWN()   (MDIO_pinValueGetPinValue(BUTTON_PORT, MOVE_DOWN))
#define READ_LEFT()   (MDIO_pinValueGetPinValue(BUTTON_PORT, MOVE_LEFT))
#define READ_RIGHT()  (MDIO_pinValueGetPinValue(BUTTON_PORT, MOVE_RIGHT))
#define READ_RESET()  (MDIO_pinValueGetPinValue(BUTTON_PORT, RESET))

#define IS_PRESSED(current, previous) ((current == DIO_LOW) && (previous == DIO_HIGH))
```

```c
    g_prevUp = currentUp;
    g_prevDown = currentDown;
    g_prevLeft = currentLeft;
    g_prevRight = currentRight;
    g_prevReset = currentReset;
}
```

snakeUpdate

snakeUpdate() updates the snake after receiving the player's input. It changes the snake's position according to its current direction and updates the snake body.

```
    //checks if head at same position of food
    u8 foodEaten =(newHead.x == g_food.x && newHead.y == g_food.y);

    if(checkCollision(newHead))
    {
        g_snake.isDead = 1;
        g_gameOver = 1;
        return;
    }

    //shifts position data of body parts in the direction of the tail
    for(u8 i = g_snake.length - 1; i > 0; i--)
    {
        g_snake.body[i] = g_snake.body[i - 1];
    }

    g_snake.body[0] = newHead;

    if(foodEaten)
    {
        HBuzzer_voidBeep();

        if(g_snake.length < MAX_SNAKE_LENGTH)
        {
            g_snake.length++;

            g_snake.body[g_snake.length - 1] =
                g_snake.body[g_snake.length - 2];
        }

        g_snake.score += 5;
        SevenSegment_DisplayNumber((u8)g_snake.score);

        generateFood();
    }
}
```

```
void snakeUpdate(void)// updates snake data
{
    if(g_snake.isDead || g_gameOver)
    {
        return;
    }


    g_snake.currentDir = g_snake.nextDir;


    Position newHead = g_snake.body[0]; //initialization


    switch(g_snake.currentDir)
    {
        case DIR_UP:
            newHead.y--;
            break;

        case DIR_DOWN:
            newHead.y++;
            break;

        case DIR_LEFT:
            newHead.x--;
            break;

        case DIR_RIGHT:
            newHead.x++;
            break;

        default:
            break;
    }
```

snakeDraw

snakeDraw() is responsible for displaying the current snake on the LCD. After the snake position is updated, this function draws the snake in its new position.

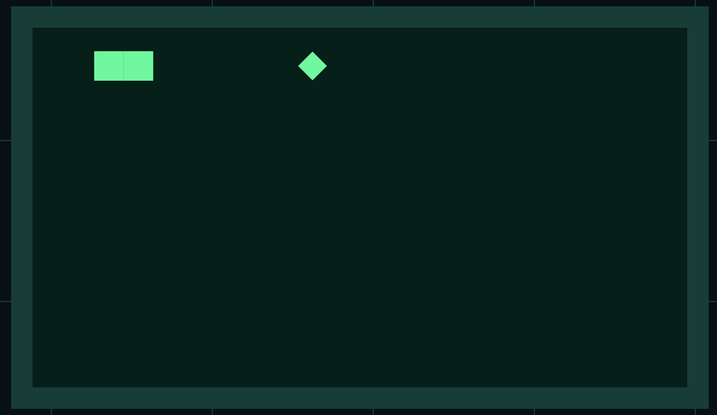
```
void snakeDraw(void)
{
    //clears LCD quickly

    for(u8 row = 0; row < 4; row++)
    {
        for(u8 col = 0; col < 20; col++)
        {
            g_displayBuffer[row][col] = ' ';
        }
    }
```

```
    // assign position of snake to LCD buffer

    for(u8 i = 0; i < g_snake.length; i++)
    {
        u8 x = g_snake.body[i].x;
        u8 y = g_snake.body[i].y;
        u8 symbol;

        if(i == 0)
        {
            switch(g_snake.currentDir)
            {
                case DIR_UP:    symbol = HEAD_UP; break;
                case DIR_DOWN:  symbol = HEAD_DOWN; break;
                case DIR_LEFT:  symbol = HEAD_LEFT; break;
                case DIR_RIGHT: symbol = HEAD_RIGHT; break;
            }
            g_displayBuffer[y][x] = symbol;
        }
        else
        {
            g_displayBuffer[y][x] = SOLID_BLOCK;
        }
    }

    g_displayBuffer[g_food.y][g_food.x] = DIAMOND;
```

```
    //send data to LCD
    for(u8 row = 0; row < 4; row++)
    {
        HLCD_voidGoToPos(row + 1, col1);

        for(u8 col = 0; col < 20; col++)
        {
            HLCD_voidSendData(g_displayBuffer[row][col]);
        }
    }

    // Keep displaying the current score
    SevenSegment_DisplayNumber((u8)g_snake.score);
}
```

Collision Detection Logic

```
u8 checkCollision(Position newHead)
{
    // check if snake is going to hit the borders

    if(newHead.x < 0 ||newHead.x >= 20 ||newHead.y < 0 ||newHead.y >= 4)
    {
        return 1;
    }

    // check if snake is going to hit its own body
    for(u8 i = 0; i < g_snake.length - 1; i++)
    {
        if(newHead.x == g_snake.body[i].x && newHead.y == g_snake.body[i].y)
        {
            return 1;
        }
    }

    return 0;
}
```



WALL COLLISION

SELF COLLISION

FOOD COLLISION

checkCollision(Position head)

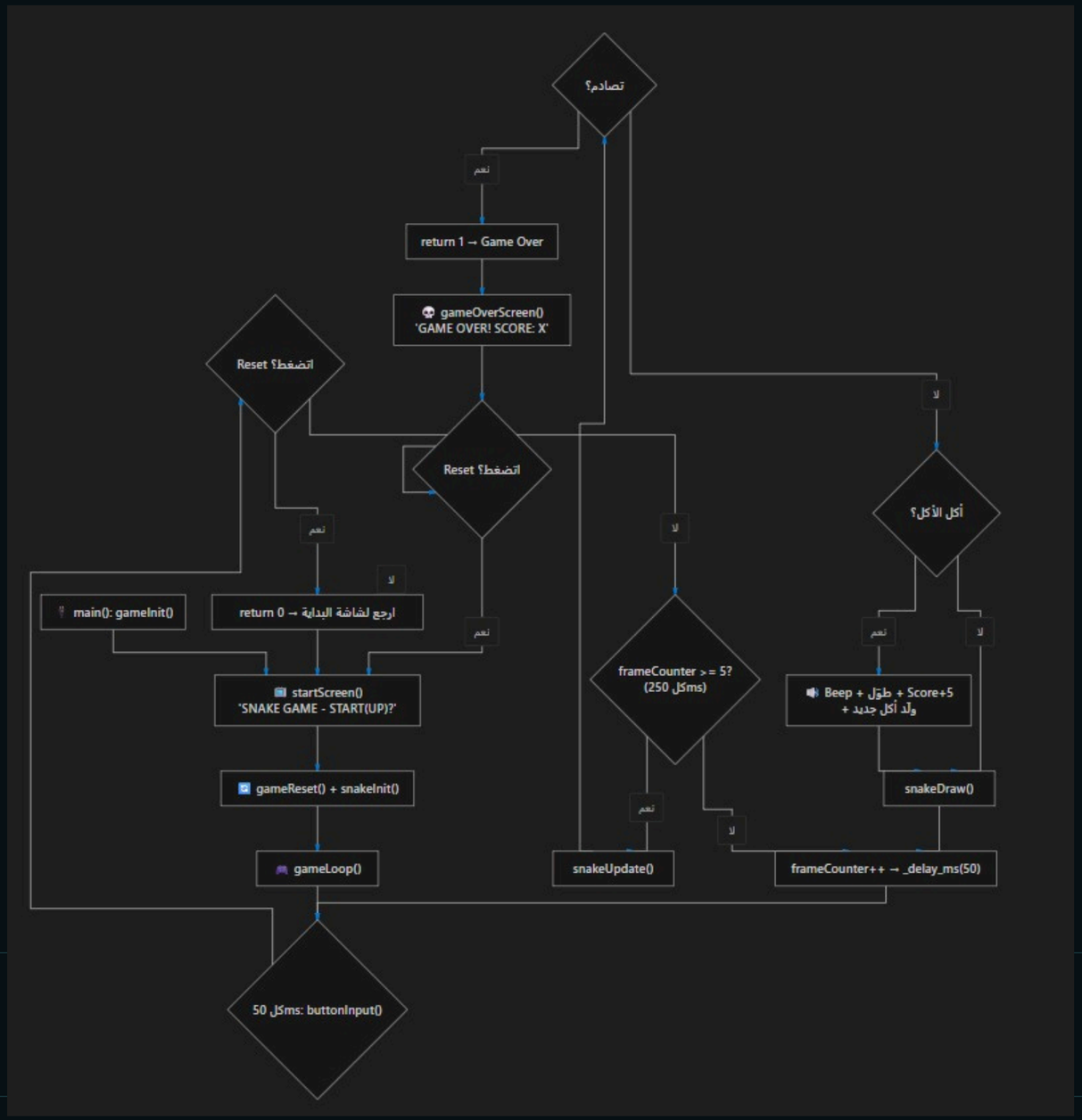
if (head.x < 0 || head.x >= WIDTH || head.y < 0 || head.y >= HEIGHT) return true;
if (snakeBodyContains(head)) return true;
return false;

checkCollision() checks whether the snake has collided with something. First, it checks whether the head is outside the game boundaries. Then, it checks whether the head is inside the snake's own body. If a collision happens, the function returns true; otherwise, it returns false.

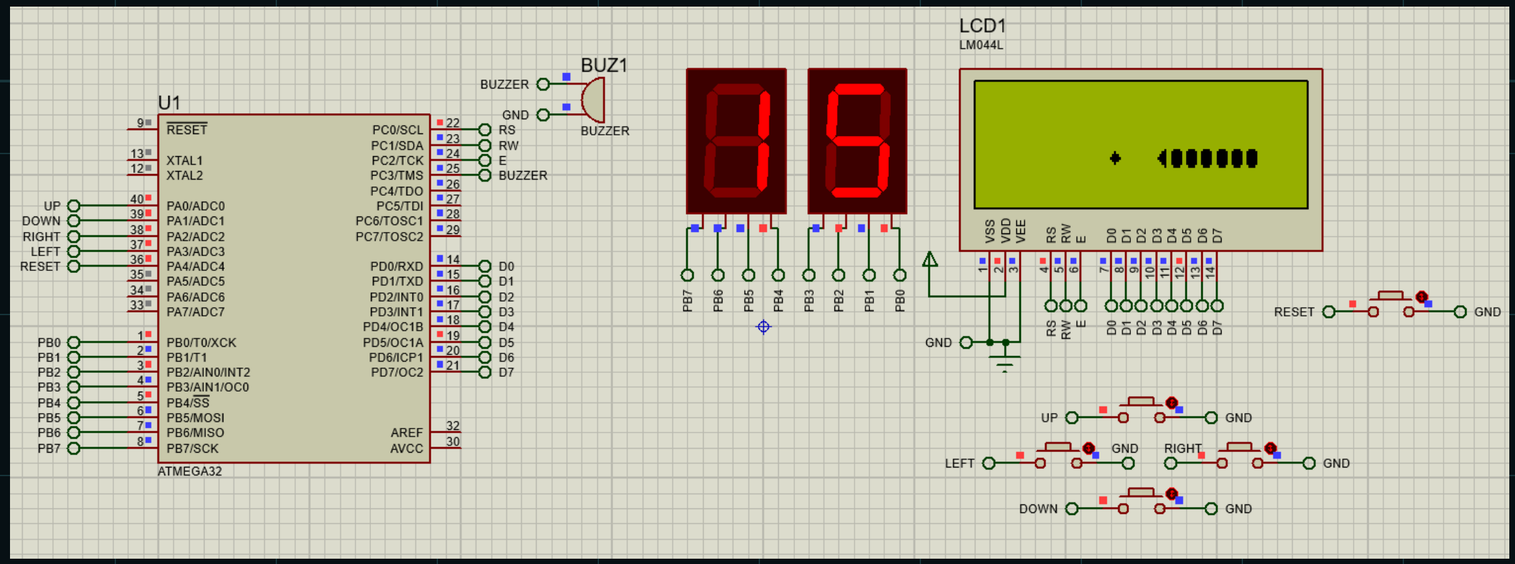
gameLoop

gameLoop() controls the main gameplay. It repeatedly handles the player's input, updates the snake, checks for collisions, draws the updated game, and handles the game-over condition

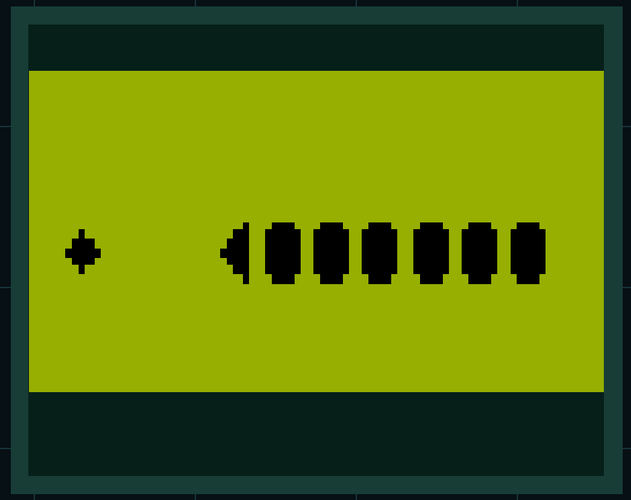
```
u8 gameLoop(void)
{
    u8 frameCounter = 0;//counter to work with different speeds

    while(g_gameRunning) {
        buttonInput();//is checked every 50ms

        if(!g_gameRunning) {
            return 0;
        }

        if(frameCounter >= 5) {// frames are updated every 50*5=250ms
            snakeUpdate();
            frameCounter = 0;

            if(g_snake.isDead) {
                g_gameOver = 1;
                g_gameRunning = 0;

                return 1;
            }

            snakeDraw();
        }
        frameCounter++;
        _delay_ms(50);
    }
}
```


تصادم؟
نعم
return 1 → Game Over
gameOverScreen()
'GAME OVER! SCORE: X'
Reset اتضغط؟
Reset اتضغط؟
لا
أكل الأكل؟
نعم
لا
main(): gameInit()
return 0 → ارجع لشاشة البداية
نعم
لا
frameCounter >= 5?
(250 كل ms)
Beep + طوّل Score+5
ولّد أكل جديد +
نعم
لا
startScreen()
'SNAKE GAME - START(UP)?'
gameReset() + snakeInit()
snakeDraw()
نعم
لا
gameLoop()
snakeUpdate()
frameCounter++ → _delay_ms(50)
50 كل ms: buttonInput()

Game Experience



START SCREEN



GAMEPLAY

GAME OVER

Q&A

Questions / Discussion